

SMART DISPLAY MODULE SPECIFICATION

1.85 Inch AI Smart Display with Touch	
Model:	SmartRing Plus
Version:	V2.0
Date:	2026-08-08

Customer Confirmation

Approved by	Notes

REVISION HISTORY

Revision	Date	Contents of Revision Change	Remark
V1.0	20260106	Preliminary release	
V1.1	20260121	Add version description and add a hyperlink to the datasheet	
V2.0	20260808	PCB Update	

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1. Product Overview



- ✧ **Main Control Chip:** ESP32-S3-N16R8 (dual-core MCU, integrated with Wi-Fi and Bluetooth BLE5.0, 240MHz main frequency, 520KB SRAM, 8MB PSRAM, 448KB ROM, 16MB Flash).
- ✧ **Display Screen:** 360×360 resolution, capacitive touch.
- ✧ **Module Functions:** Includes LCD display, backlight control, touchscreen control, I2S digital microphone, I2S digital-to-analog conversion, TF card interface, IMU(QMI8658A), and Rechargeable battery.
- ✧ **Development Environment:** Supports secondary development via Arduino IDE, ESP IDE, MicroPython, PlatformIO, etc., and UI development with LVGL.
- ✧ **Manufacturing Process:** CNC-machined casing, delicate and exquisite.
- ✧ **Supports Xiaozhi:** Provide the source code .

2. Product Functions

- Computer secondary screen function (AIDA64): Built-in 5 styles.
- Audio spectrum pickup function.
- MP3 music playback function: Allows users to add MP3 files and decode 320K high-quality MP3.
- Digital photo frame function: Supports user-added photos.
- MJPEG playback function: Enables users to add MJPEG files.
- Balance ball game: Equipped with a gyroscope, controlled via gyroscope sensor.
- Themed clock display function: Defaults to international time zone, adjustable to China time zone.
- Wireless power supply function (supports Qi protocol): Achieves true wireless operation when used with a wireless power bank.
- Real - time weather function: obtain weather information after connecting to the Internet via WIFI.
- Subsequent product function upgrades.
- Please transfer to the document for instructions for use:
 - TAIJI VIEWE Pi
 - VIEWE xiaozhi

3. Version control

3.1 Purpose

To standardize the version management of the **SmartRing-Plus Series products**, clarify the core differences between products with different configurations, and ensure that the products can be accurately identified and controlled in all processes including R&D, production, procurement, warehousing, and after-sales service, these version control instructions are specially formulated.

3.2 Description of Core Version Differences

The SmartRing-Plus series includes three versions, with the differences as follows:

Product Version	Matching Screen Model	Core Difference
SmartRing-Plus-A	VIEWE UE018HV-RB39-A002A	Screen model and corresponding initialization parameters
SmartRing-Plus-B	VIEWE UE018HV-RB39-A004A	Screen model and corresponding initialization parameters
SmartRing-Plus-B V2	VIEWE UE018HV-RB39-A004A	The power management chip is changed from ETA6002E8A to AXP2101.

3.3 Version Identification and Recognition Methods

Product Nameplate Identification:

The product packaging will clearly indicate the product version model (**SmartRing-Plus-A / SmartRing-Plus-B/SmartRing-Plus-B V2**) for intuitive identification.

Firmware:

- Before leaving the factory, the product shall be burned with the initialization parameter firmware matching the version, and the firmware version number suffix shall distinguish the version:
 - SmartRing-Plus-A firmware version suffix: A
 - SmartRing-Plus-B firmware version suffix: B (No distinction shall be made between Version B V2 and B.)

Software Update:

- We will release version-specific sample codes; please select the corresponding reference according to the version during secondary development.

4. Product information

4.1 Brie Information

- 1) Outline Dimension: ϕ 57.6 Round
- 2) Thickness: 12.2mm
- 3) Interaction Method: Touch
- 4) Shell Color: Metallic black/Silver/Customized
- 5) Power: DC 5V, 1A

System

- 1) OS: RTOS
- 2) CPU: ESP32-S3 240Mhz
- 3) RAM: 8MB
- 4) Flash: 16MB
- 5) Interface: UART/USB
- 6) Support 2.4GHz Wi-Fi、BLE 5、BLE Mesh

For more information on ESP32-S3-R8, please refer to the following link: [datasheet_en.pdf](#)(English) or [datasheet_cn.pdf](#)(Chinese)

Display

- 1) Size: 1.85 Inch
- 2) Resolution: 360 *360
- 3) Mode: IPS
- 4) Interface Mode: QSPI
- 5) Driver IC: ST77916 TP IC: CST816S
- 6) Brightness: 400 cd/m²

More information about Display can be found here: **SmartRing-Plus-A** [Display Specification.pdf](#), **SmartRing-Plus-B** [Display Specification.pdf](#)

Other

Audio

- 1) Model: ES8311 + NS4150B
- 2) For more information: [datasheet](#)

Power IC

- 1) Model: AXP2101
- 2) For more information: [datasheet](#)

MIC

- 1) Model: ZTS6216
- 2) For more information: [datasheet](#)

IMU

- 1) Model: QMI8658A
- 2) For more information: [datasheet](#)

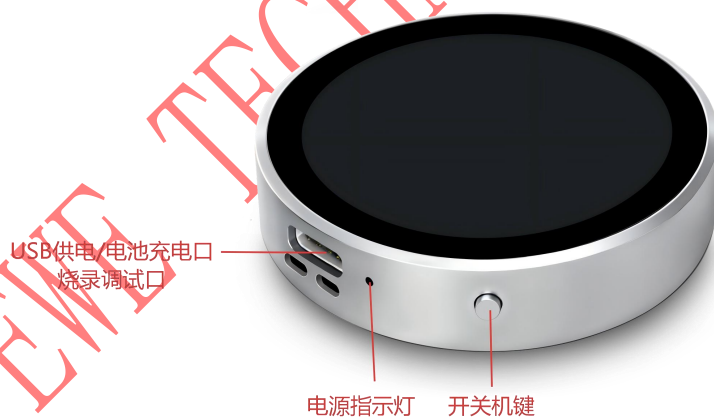
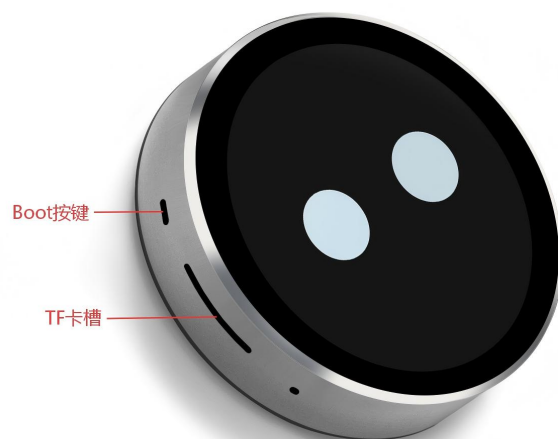
Battery

- 1) Voltage: 3.7V
- 2) Capacity: 600mAh

Horn

- 1) Impedance/Power: 8 Ω / 1W
- 2) Terminal spacing: 1.25

4.2 Exposed Interface Description



4.3 Display Information

Features	Details	Unit
Display Size(Diagonal)	1.85	inch
LCD type	α -Si TFT	-
Display Mode	IPS /Transmissive / Normally Black	-
Resolution	360RGB x 360	-
Active Area	45.68(H) $\times$ 45.68(V)	mm
Module Outline	48.08(H) $\times$ 49.95(V) $\times$ 2.12(T)	mm
Display Colors	262K	-
Interface	QSPI	-
Driver IC	77916	-
Touch IC	CST816S	-
TP Viewing Area	46.08(H) $\times$ 46.08(V)	mm
TP Outline(assembly)	55(H) $\times$ 55(V) $\times$ 0.7(T)	mm
Luminance on surface	400	cd/m ²
View Direction	All	Best image
Contrast ratio	1200:1	
Color gamut	70%	
PPI	200	-
Window effect	No one black	-
Cover plate surface effect	AF	-
Operating Temperature	-20 $\sim$ 70	°C
Storage Temperature	-30 $\sim$ 80	°C
Weight	TBD	g
connector	OK-F302-39115	

4.4 PIN Information

ESP Pin NO.	FUNCTION
GPIO0	BOOT
GPIO1	BAT_ADC
GPIO2	SDMMC_D1
GPIO3	SDMMC_D0
GPIO4	SDMMC_SCK
GPIO5	SDMMC_CMD
GPIO6	SDMMC_D3
GPIO7	SDMMC_D2
GPIO8	I2C_SDA
GPIO9	I2C_SCL
GPIO10	LCD_QSPI_SCL
GPIO11	LCD_QSPI_CS
GPIO12	LCD_QSPI_D0
GPIO13	LCD_QSPI_D1
GPIO14	LCD_QSPI_D3
GPIO15	LCD_QSPI_D2
GPIO16	I2S_DO
GPIO17	I2S_WS
GPIO18	I2S_DI
GPIO19	USB_N
GPIO20	USB_P
GPIO21	I2S_BCK
GPIO38	LCD_TE
GPIO39	LCD_RST
GPIO40	TP_RST
GPIO41	TP_INT
GPIO42	IMU_INT1
GPIO43	UART0_RX
GPIO44	UART0_TX
GPIO45	PA_CTRL
GPIO46	LCD_BLK
GPIO47	PW_OFF/ESP_PMU_IRQ
GPIO48	I2S_MCLK

 boot	 BAT_ADC	 SD
 I2C	 LCD	 I2S
 USB	 Touch	 IMU
 UART0	 PA_CTRL	 Power

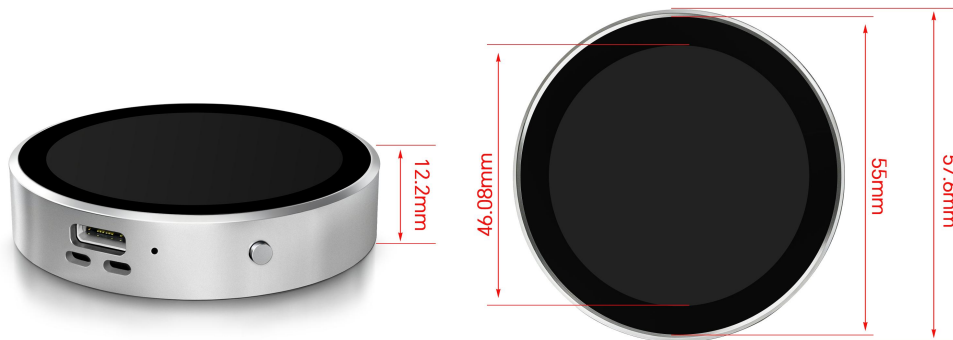
4.5 Voltage & Current

Item	Conditions	Min	Typ	Max	Unit
Power Voltage	DC	4.0	5.0	5.5	V
Operation	VCC= +5V, Maximum backlight current	-	200	-	mA
Recommended power supply:5V 1A DC					

4.6 Reliability Test

Item	Conditions	Min	Typ	Max	Unit
Working Temperature	60%RH at 5V voltage	-20	25	60	C
Storage Temperature	---	-30	25	70	C
Working Humidity	25°C	10%	60%	90%	RH
ESD	---	Contact: ±4KV Air: ±8KV			KV

5. Product Size



VIEWE TECH

6. Detailed Information

GitHub: <https://github.com/VIEWESMART/SmartRing-Plus>

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